# WEAPONS PROJECT

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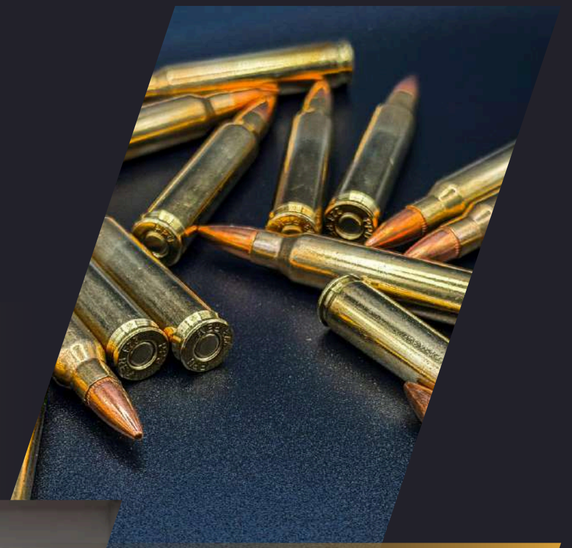
# Project Introduction

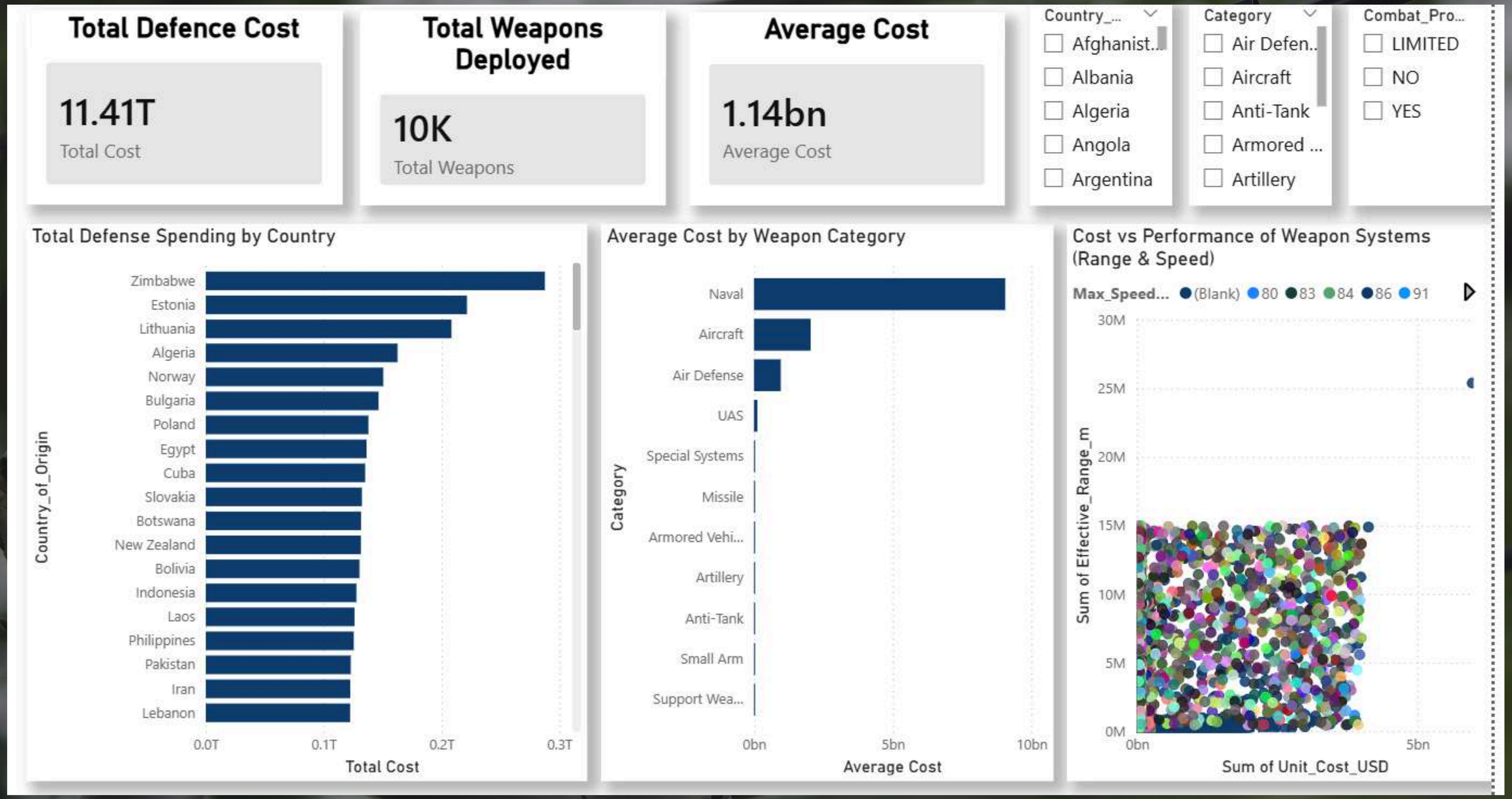
This project analyzes a global dataset of weapon systems to derive insights related to defense spending, performance efficiency, and strategic distribution. Using MySQL, the raw data was cleaned and transformed by handling missing values, standardizing formats, and validating numeric fields to create a structured analytical dataset.

Key business questions were addressed, including identifying high-investment countries, evaluating cost-to-performance efficiency, assessing manufacturer dominance, and analyzing weapon lifecycle and global adoption trends. The processed data was then visualized in Power BI Desktop to build an interactive dashboard that enables intuitive exploration of insights across multiple dimensions such as country, category, and operational characteristics.

This project demonstrates an end-to-end data analysis workflow, from data ingestion and cleaning to analysis and visualization, with a focus on delivering actionable insights for decision-making.

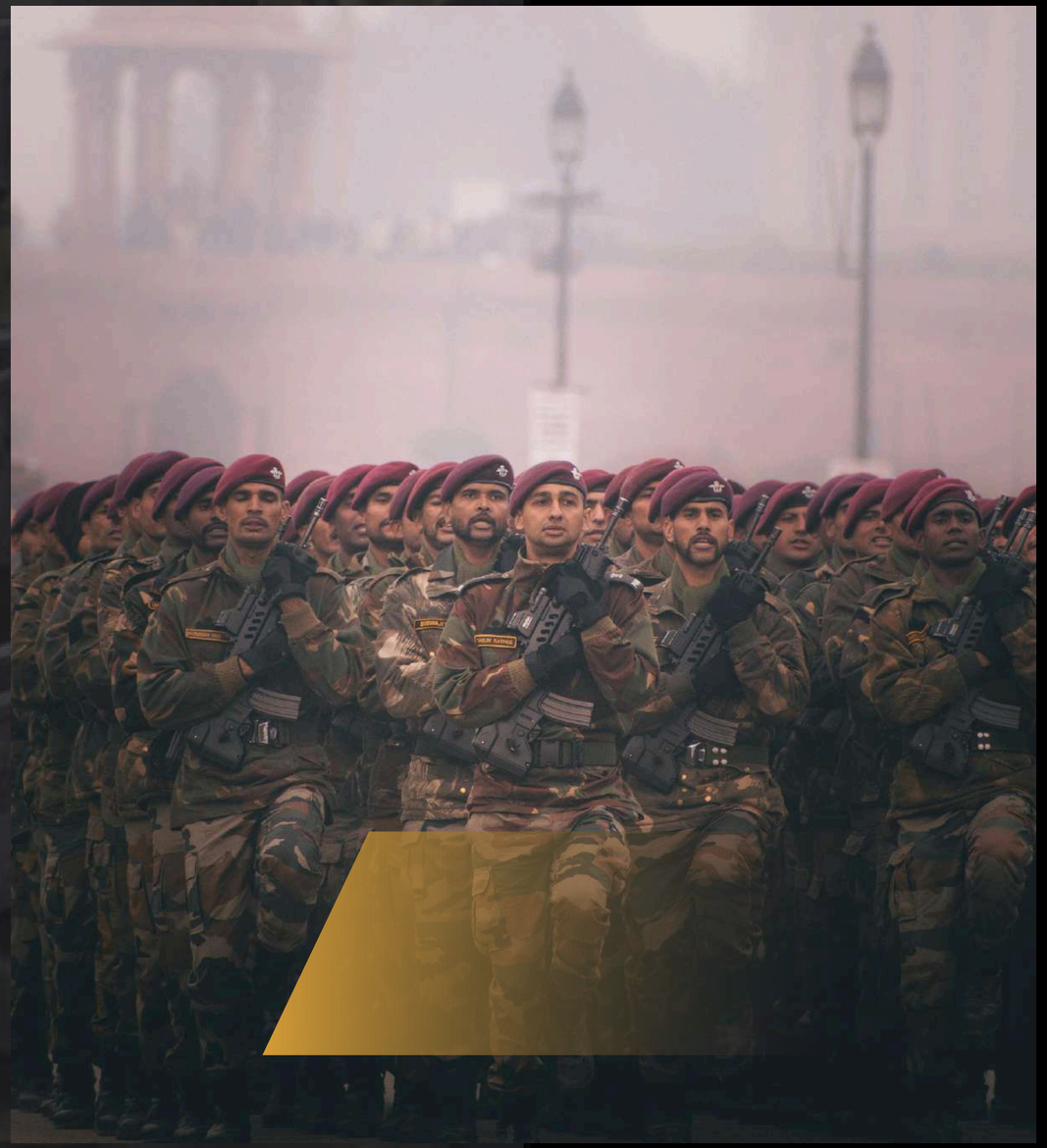


# INTERACTIVE DASHBOARD POWER BI



# Questions Answered

– Which Top 5 countries are leading in total defense investment ?
-- Which weapon categories are the most expensive?
-- Do higher-cost weapons actually perform better?
-- Which manufacturers dominate the market most?
-- Are combat-proven systems more efficient?
-- Which weapons provide the best cost-to-performance ratio?
-- Does NATO compatibility influence cost?
-- Which countries have the most diverse weapon systems?
-- What is the lifecycle of weapon systems?



## – Which Top 5 countries are leading in total defense investment ?

```sql
-- Which Top 5 countries are leading in total defense investment ?

SELECT
Country_of_Origin,
SUM(Unit_Cost_USD) AS total_spending
FROM weapons_clean
GROUP BY Country_of_Origin
ORDER BY total_spending DESC LIMIT 5;

-- ANSWER:-
-- As we can see that
-- Zimbabwe
-- Estonia
-- Lithuania
-- Algeria
-- Norway
-- are leading in total defense investment.
```

Result Grid | Filter Rows:

| Country_of_Origin | total_spending |
|---|---|
| Zimbabwe | 287738431792.00 |
| Estonia | 221621162234.00 |
| Lithuania | 208451239436.00 |
| Algeria | 162694645141.00 |
| Norway | 150591551984.00 |

## -- Which weapon categories are the most expensive?

```
-- Which weapon categories are the most expensive?

SELECT
Category,
AVG(Unit_Cost_USD) AS avg_cost
FROM weapons_clean
GROUP BY Category
ORDER BY avg_cost DESC;

-- Answer:- Naval
```

Result Grid | Filter Rows:

| Category | avg_cost |
|---|---|
| Naval | 9057285297.197838 |
| Aircraft | 2044192237.418776 |
| Air Defense | 965655965.216362 |
| UAS | 122934988.298226 |
| Special Systems | 41164435.147023 |
| Missile | 40547833.539401 |
| Armored Vehicle | 8888728.623251 |
| Artillery | 5166312.566332 |
| Anti-Tank | 361604.546667 |
| Small Arm | 139694.873638 |
| Support Weapon | 60261.049676 |

# -- Do higher-cost weapons actually perform better?

```sql
-- Do higher-cost weapons actually perform better?

SELECT
Weapon_Name,
Unit_Cost_USD,
Effective_Range_m,
Max_Speed_kmh
FROM weapons_clean
ORDER BY Unit_Cost_USD DESC;

-- Answer:- NO, by analysing the data i can say that higher-cost weapons does not actually perform better.
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

| Weapon_Name | Unit_Cost_USD | Effective_Range_m | Max_Speed_kmh |
|---|---|---|---|
| JS-207 | 17932001120.00 | NULL | NULL |
| MV-323 | 17930650840.00 | NULL | NULL |
| FS-134 | 17859860219.00 | NULL | NULL |
| USS-332 | 17857874796.00 | NULL | NULL |
| PN-468 | 17854215353.00 | NULL | NULL |
| MV-420 | 17818005483.00 | NULL | NULL |
| MV-29 | 17779642541.00 | NULL | NULL |
| PN-348 | 17744929009.00 | NULL | NULL |
| HMS-397 | 17728489071.00 | NULL | NULL |
| INS-479 | 17717041679.00 | NULL | NULL |
| PN-377 | 17697160770.00 | NULL | NULL |
| FS-389 | 17661248030.00 | NULL | NULL |
| JS-99 | 17610363391.00 | NULL | NULL |

# -- Which manufacturers dominate the market most?

```sql
-- Which manufacturers dominate the market most?

SELECT
    Manufacturer, COUNT(*) AS total_products
FROM
    weapons_clean
GROUP BY Manufacturer
ORDER BY total_products DESC;

-- Answer:- Armed Forces Ordnance.
```

| Manufacturer | total_products |
|---|---|
| Armed Forces Ordnance | 1086 |
| State Arsenal | 1070 |
| National Ordnance Plant | 1059 |
| Ministry Arms Factory | 1054 |
| State Defence Enterprise | 1051 |
| Defence Production Corp | 1028 |
| Government Defence Works | 1007 |
| National Defence Industries | 1006 |
| General Dynamics Canada | 27 |
| Polska Grupa Zbrojeniowa | 27 |
| SIG Sauer AG | 27 |
| WB Electronics | 26 |
| Chemring Norway | 25 |
| Kitron | 25 |
| Navantia | 23 |
| Thales Australia | 23 |
| EOS Defence | 22 |
| Indra Sistemas | 22 |
| Boeing Australia | 22 |
| SNC-Lavalin Defence | 21 |

## -- Are combat-proven systems more efficient?

```sql
-- Are combat-proven systems more efficient?

SELECT
    Combat_Proven,
    AVG(Unit_Cost_USD) AS avg_cost,
    AVG(Effective_Range_m) AS avg_range
FROM weapons_clean
GROUP BY Combat_Proven;

-- ANSWER:-
--  Combat-proven weapons may have:
--                                  --  higher cost
--                                  --  better reliability
```

| | Combat_Proven | avg_cost | avg_range |
|---|---|---|---|
| ▶ | YES | 1093423682.861702 | 2036846.551078 |
| | NO | 1147014281.054183 | 2153818.104571 |
| | LIMITED | 1181694007.334130 | 2153974.872972 |

## -- Which weapons provide the best cost-to-performance ratio?

```
-- Which weapons provide the best cost-to-performance ratio?

SELECT
Weapon_Name,
Unit_Cost_USD,
Effective_Range_m,
(Effective_Range_m/Unit_Cost_USD) AS efficiency_score
FROM weapons_clean
ORDER BY efficiency_score DESC;

-- ANSWER :- "RGM-397" deliver more range per dollar
```

Result Grid | Filter Rows: | Export: | Wrap Cell

| Weapon_Name | Unit_Cost_USD | Effective_Range_m | efficiency_score |
|---|---|---|---|
| RGM-397 | 162641.00 | 10802755.00 | 66.420859 |
| KEPD-Felon | 317938.00 | 12873188.00 | 40.489617 |
| DF-153 | 358520.00 | 14448888.00 | 40.301484 |
| JSM-221 | 517608.00 | 13199710.00 | 25.501364 |
| 9M-C2 | 157924.00 | 3704801.00 | 23.459392 |

# -- Does NATO compatibility influence cost?

```sql
-- Does NATO compatibility influence cost?

SELECT
    NATO_Compatible,
    AVG(Unit_Cost_USD) AS avg_cost,
    COUNT(*) AS total_systems
FROM weapons_clean
GROUP BY NATO_Compatible;

-- NATO compatible systems may cost more but they are widely adopted,
-- that means standardization increases interoperability
```

Result Grid | Filter Rows: | Export:

| NATO_Compatible | avg_cost | total_systems |
|---|---|---|
| Yes | 1059750757.461239 | 2567 |
| No | 1170208126.438927 | 2499 |
| N/A | 1203353031.728938 | 2457 |
| Partial | 1133891736.842551 | 2477 |

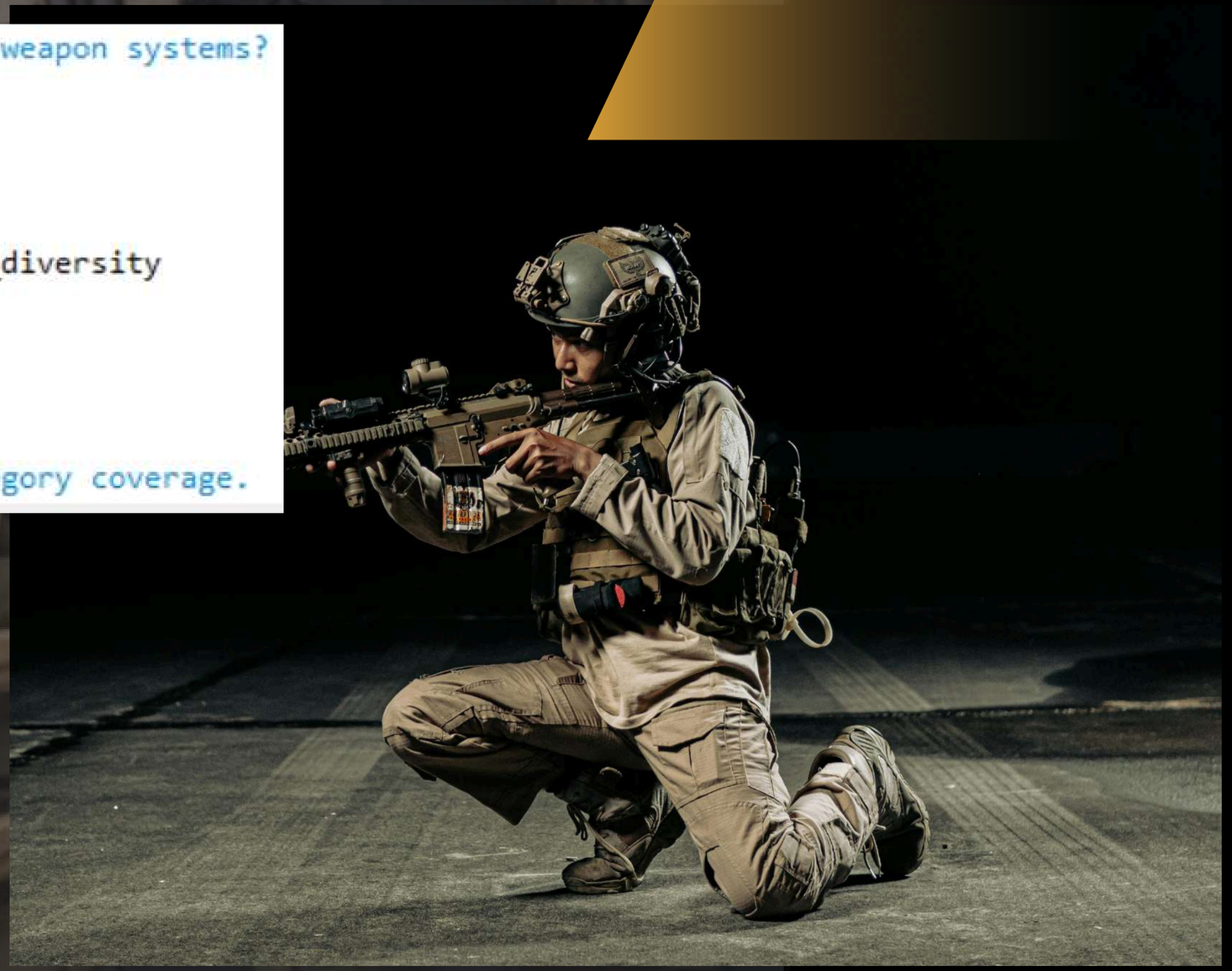
# -- Which countries have the most diverse weapon systems?

```sql
-- Which countries have the most diverse weapon systems?

SELECT
    Country_of_Origin,
    COUNT(DISTINCT Category) AS category_diversity
FROM weapons_clean
GROUP BY Country_of_Origin
ORDER BY category_diversity DESC;
-- ANSWER:- Some countries have wide category coverage.
```

Result Grid | Filter Rows:

| Country_of_Origin | category_diversity |
|---|---|
| Afghanistan | 11 |
| Albania | 11 |
| Algeria | 11 |
| Angola | 11 |
| Argentina | 11 |
| Armenia | 11 |
| Australia | 11 |
| Austria | 11 |



## -- What is the lifecycle of weapon systems?

```sql
-- What is the lifecycle of weapon systems?



SELECT
    AVG(Year_Retired - Year_Introduced) AS avg_lifecycle,
    MAX(Year_Retired - Year_Introduced) AS max_lifecycle,
    MIN(Year_Retired - Year_Introduced) AS min_lifecycle
FROM weapons_clean
WHERE Year_Retired IS NOT NULL;

-- ANSWER:- 24.7689 Years
```

Result Grid | Filter Rows:

| avg_lifecycle | max_lifecycle | min_lifecycle |
|---|---|---|
| 24.7689 | 74 | 5 |

# Thank You